

Pancreas Segmentation (512X512_256x256_128x128)

(Results for all three models (DeepLabv3+, U-Net++, SegFormer) at different resolutions)

(Framework=PyTorch, Treshold=0.40, trainig size=0.8, vaidation size=0.1, test_size=0.1)

(Train image=**938**, Test image=**118**, validation image=**117**)

(Model=DeepLabv3+)

image size 512x512 (Model=DeepLabv3+)

Number of Epochs	10	30	50
IoU	0.6081	0.6679	0.6761
Precision	0.7339	0.7919	0.7650
Recall	0.7640	0.7897	0.8460
Specificity	0.9942	0.9968	0.9954
F1 Skore	0.7271	0.7763	0.7891
Running Time (seconds)	587.47	2365.56	5862.49

image size 256x256 (Model=DeepLabv3+)

Number of Epochs	10	30	50
IoU	0.4934	0.5342	0.6183
Precision	0.7272	0.6915	0.7142
Recall	0.5561	0.6671	0.8079
Specificity	0.9968	0.9940	0.9939
F1 Skore	0.6081	0.6546	0.7427
Running Time (seconds)	410.06	1175.77	1921.20

image size 128x128 (Model=DeepLabv3+)

Number of Epochs	10	30	50
IoU	0.4730	0.5250	0.5532
Precision	0.5691	0.6131	0.6453
Recall	0.6985	0.7522	0.7775
Specificity	0.9868	0.9910	0.9907
F1 Skore	0.5962	0.6515	0.6888
Running Time (seconds)	468.61	1333.59	1866.52

(Model : U-Net++)

image size 512x512 (Model: U-Net++)

Number of Epochs	10	30	50
IoU	0.5498	0.6546	0.6675
Precision	0.5993	0.8048	0.8235
Recall	0.8208	0.7779	0.7654
Specificity	0.9882	0.9958	0.9964
F1 Skore	0.6745	0.7601	0.7765
Running Time (seconds)	732.72	3076.99	6451.02

image size 256x256 (Model: U-Net++)

Number of Epochs	10	30	50
IoU	0.6069	0.6304	0.6324
Precision	0.7336	0.7730	0.7531
Recall	0.7639	0.7610	0.7786
Specificity	0.9944	0.9958	0.9950
F1 Skore	0.7281	0.7523	0.7532
Running Time (seconds)	607.93	1581.08	2758.55

image size 128x128 (Model: U-Net++)

Number of Epochs	10	30	50
IoU	0.5384	0.5518	0.5736
Precision	0.6464	0.6884	0.7392
Recall	0.7468	0.7184	0.7004
Specificity	0.9925	0.9937	0.9960
F1 Skore	0.6628	0.6811	0.6934
Running Time (seconds)	376.13	1075.29	2677.30

(Model : SegFormer)

image size 512x512 (Model: SegFormer)

Number of Epochs	10	30	50
IoU	0.5516	0.6398	0.6831
Precision	0.6508	0.7183	0.7725
Recall	0.7749	0.8104	0.8559
Specificity	0.9895	0.9940	0.9942
F1 Skore	0.6813	0.7473	0.7997
Running Time (seconds)	1080.54	3106.61	5849.35

image size 256x256 (Model: SegFormer)

Number of Epochs	10	30	50
IoU	0.5278	0.5494	0.5901
Precision	0.6193	0.7118	0.7297
Recall	0.7317	0.6824	0.7451
Specificity	0.9903	0.9954	0.9937
F1 Skore	0.6504	0.6723	0.7140
Running Time (seconds)	403.63	948.24	1958.42

image size 128x128 (Model: SegFormer)

Number of Epochs	10	30	50 (46 e.s.)
IoU	0.4942	0.5122	0.5797
Precision	0.6805	0.6922	0.6828
Recall	0.6200	0.6293	0.7783
Specificity	0.9941	0.9956	0.9938
F1 Skore	0.6104	0.6293	0.7035
Running Time (seconds)	370.31	897.63	1736.12